

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit:

2. Subject Name: **Metamorphic Petrology Lab L-T-P: 0-0-3 Credits:**

3. Pre-requisites: **Introduction to Earth Sciences, Mineralogy, Optical Mineralogy Lab**

4. Syllabus and reference books:

Objective: The Metamorphic Petrology Laboratory Course is designed to link theoretical understanding of metamorphic processes with practical demonstrations from natural metamorphic rocks. The main emphasis of the course are the following: (1) To identify and classify different varieties of metamorphic rocks on the basis of their protolith compositions and metamorphic conditions; (2) to teach textural and micro-structural interpretations for deducing metamorphic reaction history and mineral growth in relation to specific rock deformation stages; (3) to finally link the metamorphic reaction textures and mineral assemblage evolution with qualitative P-T paths of evolution and orogenic settings.

Syllabus:

Petrographic Study of Metapelites, Metabasites and Calc-silicate Rocks in Relation to Developments of Different Generations of Deformation Fabric, Timing of Mineral Growth vis-a-vis Deformation Fabric, Identification of Metamorphic Facies, Applications of Metamorphic Reaction Textures (e.g. corona, moat, symplectites etc) to Constrain Metamorphic Reactions, with an Overall Objective to Reconstruct the Metamorphic Evolution of Orogenic Belts.

Reference Books:

- (1) Atlas of Metamorphic Rocks and their Textures by B.W.D. Yardley, W.J. Mackenzie and C.Guilford; 1990; Longman Scientific & Technical, Harlow, U.K. & J. Wiley, New York.
- (2) Microtectonics by Cees W. Passchier & Rudolph A. J. Trouw; 2005; 2nd Revised and Enlarged Edition; Springer.
- (3) Metamorphic Textures by Alan Spry; 1969; Pergamon Press.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Metamorphic Textures & Microstructures: General Introduction	6 hrs
2.	Textural and Microstructural Studies of Pelitic schists under greenschist and amphibolite facies.	12 hrs
3.	Textural and Microstructural Studies of metabasic rocks under greenschist, amphibolite and blueschist facies.	9 hrs
4.	Corona and Symplectite Textures in Pelitic, Mafic and Calc-silicate granulites	9 hrs
Total number of hours		36 hrs